# Guying Lin

(+1) 412-909-9613 | guyingl@andrew.cmu.edu | carrie-lin.github.io

## Education

**Carnegie Mellon University** — Pittsburgh, PA, USA
*Ph.D. in Computer Science* — *Sept. 2024 – Present*
- **Advisor:** Prof. **Minchen Li**

**The University of Hong Kong** — Hong Kong, China
*M.Phil. in Computer Science* — *Sept. 2022 – Aug. 2024*
- **Advisors:** Prof. **Wenping Wang** and Prof. **Taku Komura**
- Recipient of HKU Postgraduate Scholarship (Full Funding)

**Zhejiang University** — Hangzhou, China
*B.Eng. in Digital Media Technology* — *Sept. 2018 – July 2022*
- **Chu Kochen Honors College** (Elite undergraduate program of ZJU)
- **GPA:** 3.94 / 4.00 **Ranking:** Top 5%

## Experience

**Genesis AI** — May 2026 – August 2026
*Research Intern*
- **Host: Johnson Wang**

**NVIDIA** — May 2025 – May 2026
*Research Intern, Spatial Intelligence Lab*
- **Hosts: Donglai Xiang** and **Nicholas Sharp**

## Publications

1. **CoDimRecon: Agentic Reconstruction of Sim-Ready 3D Scenes with Deformable Curves, Surfaces, and Volumes**
Shuzhao Xie, Lelin Wang, **Guying Lin**, Zhi Wang, Minchen Li.
*Under Submission*, 2026.

2. **PAT3D: Physics-Augmented Text-to-3D Scene Generation**
**Guying Lin**, Kemeng Huang, Michael Liu, Ruihan Gao, Hanke Chen, Lyuhao Chen, Beijia Lu, Taku Komura, Yuan Liu, Jun-Yan Zhu, Minchen Li.
*International Conference on Learning Representations (ICLR)*, 2026.

3. **NeuVAS: Neural Implicit Surfaces for Variational Shape Modeling**
Pengfei Wang, Qiujie Dong, Fangtian Liang, Hao Pan, Lei Yang, Congyi Zhang, **Guying Lin**, Caiming Zhang, Yuanfeng Zhou, Changhe Tu, Shiqing Xin, Alla Sheffer, Xin Li, Wenping Wang.
*ACM Transactions on Graphics (TOG)*, 2025.

4. **NeuPPS: Neural Piecewise Parametric Surfaces**
Lei Yang, Yongqing Liang, Xin Li, Congyi Zhang, **Guying Lin**, Alla Sheffer, Scott Schaefer, John Keyser, Wenping Wang.
*ACM Transactions on Graphics (TOG)*, 2025.

5. **Patch-Grid: An Efficient and Feature-Preserving Neural Implicit Surface Representation**
**Guying Lin***, Lei Yang*, Congyi Zhang, Hao Pan, Yuhan Ping, Guodong Wei, Taku Komura, John Keyser, Wenping Wang.
*ACM Transactions on Graphics (TOG)*, 2025.

6. **On Optimal Sampling for Learning SDF Using MLPs Equipped with Positional Encoding**
**Guying Lin***, Lei Yang*, Yuan Liu, Congyi Zhang, Junhui Hou, Xiaogang Jin, Taku Komura, John Keyser, Wenping Wang.
*IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025.

7. **Large Images Are Gaussians: High-quality Large Image Representation with 2D Gaussian Splatting**
   Lingting Zhu, **Guying Lin**, Jinnan Chen, Xinjie Zhang, Zhenchao Jin, Zhao Wang, Lequan Yu.
   *AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

8. **ProLiF: Progressively-connected Light Field Network for Efficient View Synthesis**
   Peng Wang, Yuan Liu, **Guying Lin**, Jiatao Gu, Lingjie Liu, Taku Komura, Wenping Wang.
   *Computers & Graphics*, 2024.

9. **Deformable Endoscopic Tissues Reconstruction with Gaussian Splitting**
   Lingting Zhu, Zhao Wang, Zhenchao Jin, **Guying Lin**, Lequan Yu.
   *MICCAI EARTH*, 2024.

10. **Surface Extraction from Neural Unsigned Distance Fields**
    Congyi Zhang*, **Guying Lin***, Lei Yang, Xin Li, Taku Komura, Scott Schaefer, John Keyser, Wenping Wang.
    *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

## Academic Service & Teaching

**Technical Reviewer** 2024 – Present
*SIGGRAPH, SIGGRAPH Asia, IEEE TVCG, NeurIPS*

**Teaching Assistant** 2026
*Department of Computer Science, Carnegie Mellon University*
- 15-966: Human Protocols and Interoperability

**Teaching Assistant** 2022 – 2024
*Department of Computer Science, The University of Hong Kong*
- CCST9049: From Human Vision to Machine Vision
- COMP2396: Object-oriented Programming and Java

## Invited Talks

**Lightning Talk** Jul. 2026
*SIGGRAPH 2026: Differentiable Physics for Graphics and AI Workshop*

**Invited Talk** Jul. 2026
*DigiTwin 2026: Digital Twins for Smart Cities and Construction*

## Honors & Awards

**WiGRAPH Rising Star in Computer Graphics** 2026
**Provincial Government Scholarship**, Zhejiang Province 2022
**Honored Graduate**, Zhejiang University & Chu Kochen Honors College 2022
**Scholarship for Pilotage (Outstanding Student Award)**, Three consecutive years 2019 – 2022
**ZJU First-grade Scholarship**, Zhejiang University 2021
**WangLaoJi Scholarship**, Industry-sponsored Merit Award 2020
**Mathematics Competition for College Students**, Second Class Prize (Zhejiang Province) 2019

## Skills & Interests

**Languages:** Mandarin (Native), English (Fluent; TOEFL 110)
**Technical Skills:** Python, C++, PyTorch, Blender
**Hobbies:** Sketch & Watercolor Painting, Chinese Calligraphy, Latin Dance, Cooking
**Activities:** Minister of ZJU Youth Volunteer Association, Intern journalist at Qianjiang Evening News